$^{177}_{71}$Lu $_{106}$

1 Decay Scheme

The ^{177}Lu ground state ($J^\pi = 7/2^+$) disintegrates by β^- emission ($P_{\beta^-} = 100\%$) to the ground state ($J^\pi = 7/2^-$) and to three excited levels ($J^\pi = 9/2^-$, $11/2^-$ and $9/2^+$) of ^{177}Hf.

Le ^{177}Lu *se désintègre par émission* β^- *vers le niveau fondamental du* ^{177}Hf *via trois niveaux excités.*

2 Nuclear Data

$T_{1/2}(^{177}$Lu)	:	6,6443	(9)	d
$Q^-(^{177}$Lu)	:	496,8	(8)	keV

2.1 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log ft
$\beta^-_{0,3}$	175,5 (8)	11,55 (7)	Allowed	6,15
$\beta^-_{0,2}$	247,1 (8)	0,003 (21)	Unique 1st Forbidden	9,80
$\beta^-_{0,1}$	383,8 (8)	8,99 (28)	1st Forbidden	7,34
$\beta^-_{0,0}$	496,8 (8)	79,45 (29)	1st Forbidden	6,77

2.2 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	$P_{\gamma+ce}$ (%)	Multipolarity	α_K	α_L	α_M	α_T
$\gamma_{3,2}$(Hf)	71,6425 (6)	0,326 (21)	E1+0,04%M2	0,72 (8)	0,14 (3)	0,032 (8)	0,90 (12)
$\gamma_{1,0}$(Hf)	112,95009 (36)	20,10 (27)	M1+95,6%E2	0,808 (12)	1,084 (16)	0,270 (4)	2,23 (4)
$\gamma_{2,1}$(Hf)	136,7246 (5)	0,1008 (25)	M1+91%E2	0,54 (8)	0,448 (23)	0,111 (6)	1,13 (5)
$\gamma_{3,1}$(Hf)	208,36623 (20)	11,01 (6)	E1+0,12%M2	0,047 (4)	0,0074 (9)	0,00168 (20)	0,056 (5)
$\gamma_{2,0}$(Hf)	249,6744 (6)	0,2287 (14)	E2	0,0905 (13)	0,0375 (6)	0,00911 (13)	0,1395 (20)
$\gamma_{3,0}$(Hf)	321,3162 (6)	0,2201 (22)	E1+5,4%M2	0,040 (7)	0,0073 (13)	0,0017 (3)	0,050 (8)

3 Atomic Data

3.1 Hf

ω_K	:	0,951	(4)
$\bar{\omega}_L$	:	0,26	(1)
n_{KL}	:	0,829	(4)

3.1.1 X Radiations

		Energy (keV)	Relative probability
X_K			
	$K\alpha_2$	54,612	57,2
	$K\alpha_1$	55,7909	100
	$K\beta_3$	62,98	}
	$K\beta_1$	63,234	} 33
	$K\beta_5''$	63,662	}
	$K\beta_2$	64,98	}
	$K\beta_4$	65,132	} 8,8
	$KO_{2,3}$	65,316	}
X_L			
	$L\ell$	6,9631	
	$L\alpha$	7,8447 - 7,8991	
	$L\eta$	8,1408	
	$L\beta$	8,906 - 9,6084	
	$L\gamma$	10,2036 - 10,8905	

3.1.2 Auger Electrons

	Energy (keV)	Relative probability
Auger K		
KLL	42,601 - 46,007	100
KLX	51,391 - 55,784	53,4
KXY	60,15 - 65,34	7,08
Auger L	4,3 - 11,2	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)
e_{AL}	(Hf)	4,3 - 11,2	8,48 (7)
e_{AK}	(Hf)		
	KLL	42,601 - 46,007	0,279 (24)
	KLX	51,391 - 55,784	
	KXY	60,15 - 65,34	
$ec_{3,2\ T}$	(Hf)	6,2909 - 71,6351	0,154 (21)
$ec_{3,2\ K}$	(Hf)	6,2909 (8)	0,124 (14)
$ec_{1,0\ T}$	(Hf)	47,5992 - 112,9434	13,88 (26)
$ec_{1,0\ K}$	(Hf)	47,59925 (36)	5,03 (8)
$ec_{2,1\ T}$	(Hf)	71,3734 - 136,7176	0,0535 (24)
$ec_{1,0\ L}$	(Hf)	101,6794 - 103,3894	6,75 (11)
$ec_{1,0\ M}$	(Hf)	110,3492 - 111,2884	1,680 (26)
$ec_{1,0\ N}$	(Hf)	112,412 - 112,933	0,390 (6)
$ec_{3,1\ T}$	(Hf)	143,0150 - 208,3592	0,58 (5)
$ec_{3,1\ K}$	(Hf)	143,0150 (7)	0,490 (42)
$ec_{2,0\ T}$	(Hf)	184,3234 - 249,6676	0,02800 (43)
$ec_{3,0\ T}$	(Hf)	255,9651 - 321,3093	0,0105 (17)
$\beta^-_{0,3}$	max:	175,5 (8)	11,55 (7)
	avg:	46,70 (23)	
$\beta^-_{0,2}$	max:	247,1 (8)	0,003 (21)
	avg:	77,15 (26)	
$\beta^-_{0,1}$	max:	383,8 (8)	8,99 (28)
	avg:	110,23 (26)	
$\beta^-_{0,0}$	max:	496,8 (8)	79,45 (29)
	avg:	147,75 (27)	

5 Photon Emissions

5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)	
XL	(Hf)	6,9631 - 10,8905	3,12 (6)	
XKα_2	(Hf)	54,612	1,555 (27)	} Kα
XKα_1	(Hf)	55,7909	2,72 (5)	}
XKβ_3	(Hf)	62,98	}	
XKβ_1	(Hf)	63,234	} 0,898 (21)	K$'\beta_1$
XKβ_5''	(Hf)	63,662	}	
XKβ_2	(Hf)	64,98	}	
XKβ_4	(Hf)	65,132	} 0,240 (7)	K$'\beta_2$
XKO$_{2,3}$	(Hf)	65,316	}	

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{3,2}$(Hf)	71,6425 (6)	0,1716 (14)
$\gamma_{1,0}$(Hf)	112,95005 (36)	6,223 (32)
$\gamma_{2,1}$(Hf)	136,7245 (5)	0,04731 (44)
$\gamma_{3,1}$(Hf)	208,3661 (2)	10,425 (35)
$\gamma_{2,0}$(Hf)	249,6742 (6)	0,2007 (12)
$\gamma_{3,0}$(Hf)	321,3159 (6)	0,2096 (14)

6 Main Production Modes

^{176}Lu(n,γ)^{177}Lu σ : 1778 (75) barns
Possible impurities: ^{177m}Lu ($T_{1/2}$ = 160,4 d)

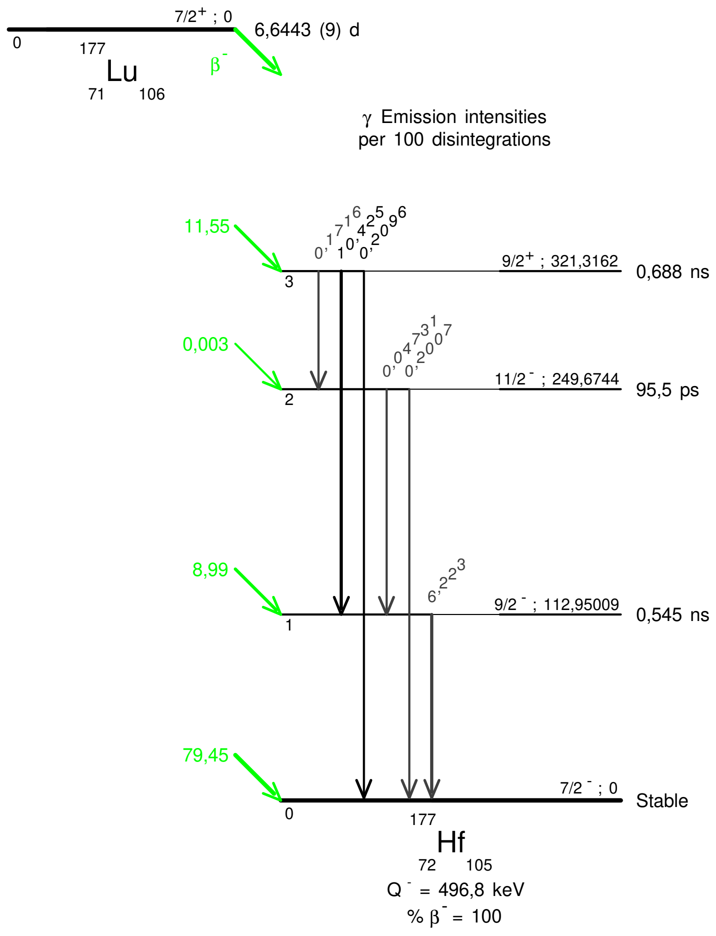
7 References

- D.G.Douglas. Phys. Rev. 75 (1949) 1960
  (Beta minus emission probability)
- P.Marmier, F.Boehm. Phys. Rev. 97 (1955) 103
  (Beta minus emission probability, Gamma-ray energies, Gamma-ray emission probabilities)
- T.Wiedling. Thesis, Univ. of Stockholm (1956)
  (Beta minus emission probability)
- R.H.Betts, O.F.Dahlinger, D.M.Munro. Can. J. Phys. 36 (1958) 73
  (Half-life)
- L.C.Schmid, W.P.Stinson. Nucl. Sci. Eng. 7 (1960) 477
  (Half-life)

- H.I.West Jr., L.G.Mann, R.J.Nagle. Phys. Rev. 124 (1961) 527
  (Gamma-ray energies, Gamma-ray emission probabilities, Mixing ratio, K ICC, L ICC)
- O.Hansen, M.C.Olesen, O.Skilbreid, B.Elbek. Nucl. Phys. 25 (1961) 634
  (Half-life)
- U.Hauser, K.Runge, G.Knissel. Nucl. Phys. 27 (1961) 632
  (Half-life)
- M.S.El-Nesr, E.Bashandy. Nucl. Phys. 31 (1962) 128
  (Beta minus emission probability)
- E.E.Berlovich, Y.K.Gusev, V.V.Ilin, M.K.Nikitin. Soviet Phys. JETP 16 (1963) 1144
  (Half-life)
- P.Alexander, F.Boehm, E.Kankeleit. Phys. Rev. 133 (1964) B284
  (Beta minus emission probability, Gamma-ray energies, Gamma-ray emission probabilities)
- D.F.Crouch, L.D.McIsaac. Report IDO-16932 (1964) 26
  (Gamma-ray emission probabilities)
- T.Novakov, J.M.Hollander. Nucl. Phys. 60 (1964) 593
  (Mixing ratio)
- B.P.Maier. Z. Physik 184 (1965) 153
  (Gamma-ray energies)
- R.Rougny, J.J.Samueli, A.Sarazin. J. Phys. (Paris) 26 (1965) 63
  (Half-life)
- A.J.Haverfield, F.M.Bernthal, J.M.Hollander. Nucl. Phys. A94 (1967) 337
  (Beta minus emission probability, Gamma-ray energies, Gamma-ray emission probabilities)
- B.V.N.Rao, S.Jnanananda. Indian J. Pure Appl. Phys. 6 (1968) 555
  (Half-life)
- S.Tornkvist, S.Strom, J.E.Thun, V.Schmidt, N.Baade. Nucl. Phys. A117 (1968) 336
  (Mixing ratio)
- B.Hrastnik, I.Basar, M.Diksic, K.Ilakovac, V.Kos, A.Ljubcic. Z. Physik 239 (1970) 25
  (Mixing ratio)
- J.F.Emery, S.A.Reynolds, E.I.Wyatt, G.I.Gleason. Nucl. Sci. Eng. 48 (1972) 319
  (Half-life)
- S.Hogberg, R.Jadrny, S.-E.Karlsson, G.Malmsten, O.Nilsson. Z. Physik 254 (1972) 89
  (Mixing ratio)
- L.Holmberg, V.Stefansson, J.Becker, C.Bargholtz, L.Gidefeldt. Phys. Scr. 6 (1972) 177
  (Mixing ratio)
- V.N.Grigoriev, D.M.Kaminker, Y.V.Sergeenkov. Bull. Acad. Sci. URSS 36 (1973) 762
  (K ICC, L ICC)
- L.M.Dautov, Y.A.Lysikov, U.M.Makhanov, Y.K.Shubnyi. Bull. Acad. Sci. URSS 36 (1973) 2210
  (Half-life)
- M.S.El-Nesr, E.Bashandy. Z. Naturforsch. 28a (1973) 1959
  (Mixing ratio)
- K.P.Gopinathan, A.P.Agnihotry, S.B.Patel, M.S.Bidarkundi. J. Phys. Soc. Jap. 34 (1973) Suppl.
  (Half-life)
- K.S.Krane, C.E.Olsen, W.A.Steyert. Phys. Rev. C10 (1974) 825
  (Mixing ratio)
- A.P.Agnihotry, K.P.Gopinathan, H.C.Jain. Phys. Rev. C9 (1974) 336
  (Gamma-ray emission probabilities, Mixing ratio, K ICC, L ICC)
- M.S.El-Nesr, E.Bashandy. Z. Naturforsch. 29a (1974) 593
  (Mixing ratio)
- B.D.Jeltema, F.M.Bernthal. Phys. Rev. C10 (1974) 778
  (K ICC, L ICC)
- H.E.Keus, W.J.Huiskamp. Physica 85B (1977) 137
  (Mixing ratio)
- V.Hnatowicz. Czech. J. Phys. B31 (1981) 260
  (Gamma-ray energies, Gamma-ray emission probabilities)
- F.Lagoutine, J.Legrand. Int. J. Appl. Radiat. Isotop. 33 (1982) 711
  (Half-life)
- D.Mehta, B.Chand, S.Singh, M.L.Garg, N.Singh, T.S.Cheema, T.N.Trehan. Nucl. Instrum. Methods Phys. Res. A260 (1987) 157
  (Gamma-ray emission probabilities)

- S.Matsui, H.Inoue, Y.Yoshizawa. Nucl. Instrum. Methods Phys. Res. A281 (1989) 568
(Gamma-ray energy)
- A.Abzouzi, M.S.Antony, A.Hachem, V.B.Ndocko Ndongue. J. Radioanal. Nucl. Chem. 144 (1990) 359
(Half-life)
- C.C.Dey, B.K.Sinha, R.Bhattacharya. Nuovo Cim. 105A (1992) 1307
(Mixing ratio)
- E.Schönfeld, H.Janssen. Nucl. Instrum. Methods Phys. Res. A369 (1996) 527
(Atomic Data)
- I.Alfter, E.Bodenstedt, W.Knichel, J.Schuth, R.Vianden. Z. Physik A355 (1996) 363
(Half-life)
- E.Schönfeld, G.Rodloff. Report PTB-6.11-98-1 (1998)
(X-ray energies)
- E.Schönfeld, G.Rodloff. Report PTB-6.11-1999-1 (1999)
(X-ray energies)
- E.Schönfeld, H.Janssen. Appl. Rad. Isotopes 52 (2000) 595
(Atomic probabilities)
- B.E.Zimmerman, M.P.Unterweger, J.W.Brodack. Appl. Rad. Isotopes 54 (2001) 623
(Half-life)
- U.Schötzig, H.Schrader, E.Schönfeld, E.Gunther, R.Klein. Appl. Rad. Isotopes 55 (2001) 89
(Beta minus emission probability, Half-life, Gamma-ray emission probabilities)
- E.Schönfeld, U.Schötzig. Appl. Rad. Isotopes 54 (2001) 785
(X-ray emission probabilities)
- M.P.Unterweger. Appl. Rad. Isotopes 56 (2002) 125
(Half-life)
- I.M.Band, M.B.Trzhaskovskaya, C.W.Nestor Jr., P.O.Tikkanen, S.Raman. At. Data Nucl. Data Tables 81 (2002) 1
(ICC)
- F.G.Kondev. Nucl. Data Sheets 98 (2003) 801
(Evaluation)
- H.Schrader. Appl. Rad. Isotopes 60 (2004) 317
(Half-life)
- T.Kibédi, T.W.Burrows, M.B.Trzhaskovskaya, P.M.Davidson, C.W.Nestor Jr. Nucl. Instrum. Methods A589 (2008) 202
(ICC)
- L.Canella, et al.. J. Radioanal. Nucl. Chem. 276 (2008) 813
(Half-life)
- M.S.Dias, F.F.V.Silva, M.F.Koskinas. Appl. Rad. Isotopes 68 (2010) 1349
(Gamma-ray intensity)
- S.Pommé, J.Paepen, T.Altzitzoglou, R.Van Ammel, E.Yeltepe. Appl. Rad. Isotopes 69 (2011) 1267
(Half-life)
- S.Deepa, K.Vijay Sai, R.Gowrishankar, D.R.Rao, K.Venkataramaniah. Appl. Rad. Isotopes 69 (2011) 869
(ICC, Gamma-ray energies and intensities)
- S.Zhu, et al.. Nucl. Instrum. Methods A652 (2011) 231
(Half-life)
- S.Deepa, K.Vijay Sai, R.Gowrishankar, K.Venkataramaniah. Eur. Phys. J. A48 (2012) 126
(ICC, Gamma-ray energies)
- K.Kossert, O.J.Nähle, O.Ott, R.Dersch. Appl. Rad. Isotopes 70 (2012) 2215
(Half-life, Gamma-ray intensities)
- R.Fitzgerald. J. Res. Natl. Inst. Stand. Technol. 117 (2012) 80
(Half-life)
- E.A.Rezende, A.R.Correia, A.Iwahara, C.J.da Silva, L.Tauhata, R.Poledna, R.L.da Silva, E.M.de Oliveira, A.E.de Oliveira. Appl. Rad. Isotopes 70 (2012) 2081
(Half-life)
- N.D'Olympia, S.Lakshmi, P.Chowdhury, E.G.Jackson, J.Glodo, K.Shah. Nucl. Instrum. Methods A728 (2013) 31
(Half-life)
- P.Dryàk, J.Sochorová, J.Solc, P.Auerbach. Appl. Rad. Isotopes 109 (2016) 160
(Half-life)
- A.Luca, M.Sahagia, M.-R.Ioan, A.Antohe, B.Luminita Savu. Appl. Rad. Isotopes 109 (2016) 146
(Half-life)

- M.A.Kellett. Appl. Rad. Isotopes 109 (2016) 129
  (Evaluation)
- K.M.Ferreira, S.M.Collins, A.J.Fenwick. EPJ Web Conf. 146 (2017) 08002
  (Half-life)
- F.G.Kondev. Nucl. Data Sheets 159 (2019) 1
  (Evaluation)
- L.Knafla, et al.. Phys. Rev. C102 (2020) 054322
  (Half-life)
- M.Wang, W.J.Huang, F.G.Kondev, G.Audi, S.Naimi. Chin. Phys. C45 (2021) 030003
  (Q-value)
- X.Mougeot. Appl. Rad. Isotopes 201 (2023) 111018
  (Beta decay calculation)
- J.Marganiec-Galazka, M.Czudek, E.Lech, A.Listkowska, P.Saganowski, Z.Tyminski, T.Ziemek. Appl. Rad. Isotopes 197 (2023) 110829
  (Half-life)
- M.-C.Lépy, B.Boyer, S.Pierre, A.Utepov, C.Bobin, L.Chambon. Appl. Rad. Isotopes 226 (2025) 112212
  (Gamma-ray emission probabilities)


7/2+ ; 0
6,6443 (9) d
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177
Lu
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β-
γ Emission intensities
per 100 disintegrations
11,55
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8,99
79,45
9/2+ ; 321,3162
0,688 ns
11/2 - ; 249,6744
95,5 ps
9/2 - ; 112,95009
0,545 ns
7/2 - ; 0
Stable
177
Hf
72 105
Q - = 496,8 keV
% β- = 100